[NOT MEASUREMENT SENSITIVE]
A-A-50427
June 26, 1996

COMMERCIAL ITEM DESCRIPTION

CLEANING COMPOUND, SOLVENT, 1,1-DICHLORO-1-FLUOROETHANE, HCFC-141B, FOR OXYGEN SYSTEMS

The General Services Administration has authorized
the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for 1,1-dichloro-1-fluoroethane cleaning compound solvent for use in oxygen system cleaning.

2. SALIENT CHARACTERISTICS.

2.1 Material. The cleaning compound shall be a solvent of 1,1-dichloro-1-fluoroethane (Chemical Abstract Service [CAS] Registry Number 1717-00-6, hydrochlorofluorocarbon, HCFC-141b).

2.2 Chemical and physical properties. The solvent cleaning compound shall conform to the requirements of table I.

2.3 Material safety data sheets. A material safety data sheet shall be prepared and submitted in accordance with FED-STD-313, meeting the requirements of Code of Federal Regulation 29 CFR 1910.1200. Questions pertinent to the effect of the cleaner on the health of personnel when used for its intended purpose shall be referred by the acquiring activity to the appropriate medical service who shall act as adviser to the acquiring activity.

2.4 Shelf life. The 1,1-dichloro-1-fluoroethane shall meet the specified requirements in table I after storage in unopened containers for not less than one year.

2.5 Product appearance. The solvent shall be a clear, colorless liquid (water white) with no evidence of particles. The appearance shall be determined by visually examining 500 mL of solvent for clarity, color, and solids in a clear container against a white and black background. The inspection shall be performed by a person with normal visual acuity, natural or corrected,

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FSC 6850

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under bright light of not less than 100 foot-candles. A general purpose flashlight with 2 fresh batteries and a fresh lamp, positioned 18 inches perpendicular to a surface will illuminate that surface with an intensity of not less than 100 foot-candles.

TABLE I. Chemical and physical properties.

Property	Requirement	Test Procedure 1/
Boiling point @ 760 mm barometric pressure	32.0 ±0.2° C (89.6 ±0.4° F)	ASTM D1078
Chemical purity, % min by weight	99.9	ASTM D3447
Impurities	Other halogenated solvents	ASTM D3447
Moisture content, ppm, max, by wt.	10	ASTM D3401
Chloride ion, ppm, max, by wt.	0.1	ASTM D3443
Nonvolatile residue, ppm, max, by wt.	5	ASTM D2109 or ASTM F331
Acid number, mg KOH/gm sample, max	0.003	ASTM D3444
Particle matter, number of particles greater than 50 microns	None	See 2.5

1/ Contractor test procedures equivalent to those specified may be substituted.

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2.6 Safety warning labels. Containers shall be marked with warning labels as follows:

DICHLOROFLUOROETHANE
HCFC-141b

WARNING

Eye contact with vapor causes irritation. Contact with liquid can cause chemical burns. Prolonged/repeated skin contact may cause dryness/irritation. Exposure can also result in freezing of skin tissue (frost bite).
Over-exposure to vapor may cause respiratory system irritation, headache, dizziness and central nervous system (CNS) depression.
Exposure to higher levels may also result in increasing sensitivity of the heart to adrenaline. Cardiac arrhythmia could result.
Exposure to very high levels could result in death due to dilution/displacement of available oxygen, resulting in death by asphyxiation. Low areas are especially susceptible to oxygen displacement. Taste/odor do not provide adequate warning of high vapor levels.
Do not use in the general vicinity of welding, open flames or hot surfaces. This could result in the decomposition of this material to irritating and toxic gases and carbonyl halides.

In case of spills/significant release to work space, warn others and leave immediately.

Use smallest quantity possible to properly perform the operation.

Use only with adequate ventilation . Refer to Material Safety Data Sheet for additional information and required precautions for use. Consult medical department authority for guidance in control and proper use of this material, and specific requirements and methods for monitoring vapor levels in the workspace.

Have an emergency escape breathing device (EEBD) immediately available for any operation where overexposure to product could occur unless immediate escape to a vapor-free area is possible. Be sure you are trained in the proper care, inspection, and use of this device.

Ensure storage areas that contain this product are properly ventilated prior to entry.

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FIRST AID

Inhalation: Immediately remove person to fresh air. If breathing has stopped, give artificial respiration. In cases of excessive exposure, pay particular attention to the potential for cardiac arrhythmia. Oxygen may be used in treating such cases if administered by medical department personnel. Use of vaso-pressor drugs, for example epinephrine (adrenaline), is inadvisable because these drugs may induce cardiac arrhythmia. If high temperature decomposition products may have been inhaled, observe the patient for immediate or delayed signs of pulmonary edema. In any of these cases, obtain prompt medical attention.

NOTE TO PHYSICIANS: Overexposure to dichlorofluoroethane has caused cardiac sensitization to epinephrine. Cardiac arrhythmia, including ventricular fibrillation could occur if epinephrine or one of its congeners is administered to patients exposed to high concentrations of dichlorofluoroethane. Medical use of epinephrine or any of its congeners is contra-indicated except for patients with no arterial profusion.

Skin contact: If pain or freezing of tissue occurs, seek prompt medical attention. If no freezing of the skin is apparent, wash areas with warm water. If material soaks through clothing, remove clothing and wash skin with soap or mild detergent and water. If irritation persists after washing, obtain medical attention. Do not re-use contaminated clothing until it is thoroughly cleaned to remove the material.

Eye contact: Immediately wash with large amounts of potable water for a minimum of 15 minutes. Avoid use of any other agents such as oil or saline solutions in the eye. Use of any other agent or dressing should only be administered by the cognizant medical department representative.

Contact lens should not be worn when using this chemical.

Ingestion: Do not induce vomiting. Seek medical attention.

2.7 Delivery options. The 1,1-dichloro-1-fluoroethane may be delivered in 5 or 55 gallon drums, or in bulk containers (tank truck, tank railroad car). that have been precleaned. Precleaning includes all caps for the containers.

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3. REGULATORY REQUIREMENTS.

3.1 Environmental considerations. The product delivered under this CID shall be in accordance with the environmental requirements of Code of Federal Regulation 40CFR.

3.2 Transportation. The transportation of the product delivered under this CID shall be in accordance with Code of Federal Regulations 49 CFR 171-190.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.1.1 Derivation of substantiating evidence of compliance. The contractor is requested to derive substantiating evidence of compliance using the following sampling:

a. Sampling 5 and 55 gallon containers:, Remove a 1-liter sample for testing in accordance with table I.

b. Sampling bulk containers for testing in accordance with table I:

(1) When the container is equipped with a pumping system that will completely recirculate its contents within one hour, take samples at intervals of one and two hours after initiation of recirculation.
(2) When the contents of the container cannot be thoroughly circulated, take three samples in accordance with ASTM-E300 representing the top, middle, and bottom sections.

4.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this commercial item description:

a. The company furnishing the solvent cleaning compound must have been producing a product meeting the requirements of this commercial item description for at least 6 months.

b. The company must have offered the solvent cleaner to a government laboratory for inspection.

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5. PACKAGING.

5.1 Preservation, packing and marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Copies of the Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, US Government Printing Office, Washington DC 20402.

6.1.2 Federal Government activities may obtain copies of Federal Standard Nos. 313 and 709 from established distribution points within their agencies. Other activities may obtain the standards for bidding purposes from the General Services Administration, Federal Supply Service Bureau, Specification section, Suite 8100, 479 L'Enfant Plaza, SW, Washington, DC 20407.

6.1.3 The following test methods are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19103:

ASTM-D1078	Test Method for Distillation Range of Volatile Organic Liquids
ASTM-D2109	Test Method for Nonvolatile Matter in Halogenated Organic Solvents and their Admixtures
ASTM-D3401	Test Method for Water in Halogenated Organic Solvents and their Admixtures
ASTM-D3443	Test Method for Chloride in Trichlorotrifluoroethane,
ASTM-D3444	Test Method for Total Acid Number of Trichlorotrifluoroethane
ASTM-D3447	Test Method for Purity of Trichlorotrifluoroethane
ASTM-E300	Practice for Sampling Industrial Chemicals
ASTM-F331	Test Method for Nonvolatile Residue of Halogenated Solvent Extract from Aerospace Components (Using Rotary Flash Evaporator)

6.2 Government laboratory. Manufacturers wanting to have their cleaning compound solvent inspected may do so by contacting the Naval Air Warfare Center Aircraft Division, Mail Stop 3, Bldg 2188, Code 4.3.4.1, 48066 Shaw Road, Patuxent River, MD 20670-5304.

6.3 Use. The use of this compound for cleaning oxygen systems should be accomplished only with approved procedures.

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MILITARY INTERESTS:

Custodians:
Army - EA
Navy - AS
Air Force - 68

Preparing Activity:

Navy - AS
DoD Project 6850-1173

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